

POLYNOMIALS

①

Polynomial is a mathematical expression of one or more algebraic terms each of which consists of a constant multiplied by one or more variable raised to a non-negative integral power such as (ax^2+bx+c)

Kinds of Polynomial : $\rightarrow$

- * A monomial is a polynomial with only one term, such as $3x$, $4xy$, 7 and $3x^2y^3$ etc
- * A binomial is a polynomial with exactly two terms such as $x+3$, $4x^2+5x$, $x+2y^7$ etc
- * A trinomial is a polynomial with exactly three terms such as x^4+3x^3-2 , $3x^2+3x+2$ etc

Another special kind of polynomial is quadratic equation which is polynomial of degree 2. It looks like ax^2+bx+c where a , b and c are constant.

Another one is Cubic polynomial. which has degree 3. It looks like ax^3+bx^2+cx+c

Some more examples

$3x^2+2 \rightarrow$ Quadratic, $3x^3+2 \rightarrow$ Cubic polynomial.

Zero polynomial : $\rightarrow$

The constant polynomial 0 is called the zero polynomial. The degree of zero polynomial is not defined.

Exercise 2.1

Q 1) Which of the following expressions are polynomials in one variable and which are not? state reason for your answer.

i) $4x^2 - 3x + 7$

Sol: $\rightarrow$ Polynomial in one variable 'x'

ii) $y^2 + \sqrt{2}$

Sol Polynomial in one variable 'y'

iii) $3\sqrt{t} + \sqrt{2}$

Sol It is not a polynomial as power of t is not a whole number.

iv) $y + \frac{2}{y}$

Sol: $\rightarrow$ This is not a polynomial as again power of y in second term is not a whole number

v) $x^{10} + y^3 + t^{50}$

Sol: $\rightarrow$ This is not a polynomial in one variable but a polynomial in 3 variables.

Q2 Write the coefficients of x^2 in each of the following:

i) $2 + x^2 + x$

Sol: $\rightarrow$ Coefficient of x^2 is 1

ii) $2 - x^2 + x^3$

Sol: $\rightarrow$ Coefficient of $x^2 = -1$

iii) $\frac{\pi}{2}x^2 + x$

Sol: $\rightarrow$ Coefficient of $x^2 = \frac{\pi}{2}$

iv) $\sqrt{2}x - 1$

Sol: $\rightarrow x^2$ is not present hence no coefficient

Q3 Give one example each of a binomial of degree 35, and of a monomial of degree 100

Sol $x^{35} + 5$ is a binomial of degree 35

$2y^{100}$ is a monomial of degree 100

Q4 Write the degree of each of the following polynomials:

i) $5x^3 + 4x^2 + 7x$

Sol: $\rightarrow$ Degree is 3 as highest power

ii) $4 - y^2$

Sol $\rightarrow$ Degree is 2

iii) $5t - \sqrt{7}$

Sol: $\rightarrow$ Degree is 1

iv) 3

Sol $\rightarrow$ Degree is 0

Q5 Classify the following as linear, quadratic and cubic polynomials:

i) $x^2 + x$

Sol: $\rightarrow$ quadratic polynomial

ii) $x - x^3 \rightarrow$ cubic polynomial

iii) $y + y^2 + 4 \rightarrow$ quadratic polynomial

iv) $1 + x \rightarrow$ linear polynomial

v) $3t \rightarrow$ linear polynomial

vi) $r^2 \rightarrow$ quadratic polynomial

vii) $7x^3 \rightarrow$ cubic polynomial

Zero of a Polynomial

A zero or root of a polynomial function is a number that when plugged in for the variable, makes the function equal to zero.

Note: $\rightarrow$ Every linear polynomial has one and only one zero

$\rightarrow$ A polynomial can have more than one zero e.g. quadratic equations have 2 zero cubic have 3 zeroes

$\rightarrow$ A zero of polynomial need not to be zero.

$\rightarrow$ 0 may be a zero of polynomial.

EXERCISE 2.2

Q① find the value of polynomial $5x - 4x^2 + 3$ at
 i) $x=0$ ii) $x=-1$ iii) $x=2$

Solⁿ: $\rightarrow$ Let $P(x) = 5x - 4x^2 + 3$

Put $x=0$

$$\therefore P(0) = 5(0) - 4(0)^2 + 3$$

$$\Rightarrow P(0) = 0 - 0 + 3$$

$\Rightarrow$ At $x=0$ value of given $P(x) = 3$

$$\text{ie } \boxed{P(0) = 3}$$

$$\begin{aligned} \text{ii) Now } P(-1) &= 5(-1) - 4(-1)^2 + 3 \\ &= -5 - 4 + 3 \\ &= -6 \end{aligned}$$

$$\therefore \boxed{P(-1) = -6}$$

$$\begin{aligned} \text{iii) } P(2) &= 5(2) - 4(2)^2 + 3 \\ &= 10 - 16 + 3 \end{aligned}$$

$$\Rightarrow \boxed{P(2) = -3}$$

Q2 Find $P(0)$, $P(1)$ and $P(2)$ for each of the following polynomials! $\rightarrow$

$$\text{i) } P(y) = y^2 - y + 1$$

$$\text{Sol } P(0) = (0)^2 - (0) + 1 = 1$$

$$P(1) = (1)^2 - (1) + 1 = 1$$

$$P(2) = (2)^2 - (2) + 1 = 3$$

$$\text{ii) } P(t) = 2 + t + 2t^2 - t^3$$

$$\begin{aligned}\text{Sol: } \rightarrow P(0) &\rightarrow 2 + 0 + 2(0)^2 - (0)^3 \\ &\rightarrow 2 + 0 + 0 - 0 = 2\end{aligned}$$

$$\begin{aligned}P(1) &= 2 + (1) + 2(1)^2 - (1)^3 \\ &= 2 + 1 + 2 - 1\end{aligned}$$

$$P(1) = 4$$

$$\begin{aligned}P(2) &= 2 + (2) + 2(2)^2 - (2)^3 \\ &= 2 + 2 + 8 - 8\end{aligned}$$

$$P(2) = 4$$

$$\text{iii) } P(x) = x^3$$

$$P(0) = (0)^3 = 0$$

$$P(1) = (1)^3 = 1$$

$$P(2) = (2)^3 = 8$$

$$\text{iv) } P(x) = (x-1)(x+1)$$

$$\begin{aligned}P(0) &= (0-1)(0+1) \\ &= (-1)(+1)\end{aligned}$$

$$P(0) = -1 \text{ Ans//}$$

$$P(1) = (1-1)(1+1)$$

$$P(1) = 0 \text{ Ans//}$$

$$P(2) = (2-1)(2+1)$$

$$P(2) = 3 \text{ Ans//}$$

(4)

Q3 Verify whether the following are zeroes of the polynomial indicated against them

i) $P(x) = 3x + 1$, $x = -\frac{1}{3}$

Sol: $\rightarrow$

$$\begin{aligned} P\left(-\frac{1}{3}\right) &= 3\left(-\frac{1}{3}\right) + 1 \\ &= -1 + 1 \\ &= 0 \quad \text{Ans//} \end{aligned}$$

ii) $P(x) = 5x - \pi$, $x = \frac{4}{5}$

$$\begin{aligned} \text{Sol: } \rightarrow P\left(\frac{4}{5}\right) &= 5\left(\frac{4}{5}\right) - \pi \\ &= 4 - \pi \quad \text{Ans//} \end{aligned}$$

iii) $P(x) = x^2 - 1$, $x = 1, -1$

$$\begin{aligned} P(1) &= (1)^2 - 1 \\ &= 1 - 1 = 0 \quad \text{Ans//} \end{aligned}$$

$$\begin{aligned} P(-1) &= (-1)^2 - (1) \\ &= 1 - 1 = 0 \quad \text{Ans//} \end{aligned}$$

iv) $P(x) = (x+1)(x-2)$, $x = -1, 2$

$$\begin{aligned} P(-1) &= (-1+1)(2-2) \\ &= 0 \quad \text{Ans//} \end{aligned}$$

$$\begin{aligned} P(2) &= (2+1)(2-2) \\ &= 3(0) \\ &= 0 \quad \text{Ans//} \end{aligned}$$

v) $P(x) = x^2$, $x = 0$

$$\begin{aligned} P(0) &= (0)^2 \\ &= 0 \quad \text{Ans//} \end{aligned}$$

$$vi) P(x) = lx + m, x = -\frac{m}{l}$$

$$\text{Sol: } \rightarrow P\left(-\frac{m}{l}\right) = l\left(-\frac{m}{l}\right) + m = -m + m = 0$$

$$vii) P(x) = 3x^2 - 1, x = -\frac{1}{\sqrt{3}}, \frac{2}{\sqrt{3}}$$

$$\begin{aligned} P\left(-\frac{1}{\sqrt{3}}\right) &= 3\left(-\frac{1}{\sqrt{3}}\right)^2 - 1 \\ &= 3 - 1 = 2 \end{aligned}$$

$$P\left(\frac{2}{\sqrt{3}}\right) = 3\left(\frac{2}{\sqrt{3}}\right)^2 - 1$$

$$3(4) - 1 = 12 - 1 = 11 \quad \text{Ans//}$$

$$viii) P(x) = 2x + 1, x = \frac{1}{2}$$

$$P\left(\frac{1}{2}\right) = 2\left(\frac{1}{2}\right) + 1 = 1 + 1 = 2 \quad \text{Ans//}$$

Q4 Find the zero of the polynomial in each of the following cases.

$$i) P(x) = x + 5$$

$$\text{Sol Put } P(x) = 0 \Rightarrow x + 5 = 0 \Rightarrow x = -5$$

$\therefore -5$ is zero of given polynomial

$$ii) P(x) = x - 5$$

$$\text{Sol: } \rightarrow \text{Put } P(x) = 0 \Rightarrow x - 5 = 0 \Rightarrow x = 5$$

$\therefore +5$ is zero of given $P(x)$

$$iii) P(x) = 2x + 5$$

$$\text{Put } P(x) = 0 \Rightarrow 2x + 5 = 0 \Rightarrow 2x = -5 \Rightarrow x = -\frac{5}{2}$$

$\therefore -\frac{5}{2}$ is zero of given $P(x)$

by $x-a$, then quotient is $q(x)$ and the remainder is $r(x)$ i.e.

$$P(x) = (x-a)q(x) + r(x)$$

As Dividend = (Divisor $\times$ Quotient) + Remainder

Since the degree of $x-a$ is 1 and the degree of $r(x)$ is less than the degree of $x-a$, the degree of $r(x) = 0$. This means that $r(x)$ is constant say r .

So for every value of x , $r(x) = r$

Therefore
$$P(x) = (x-a)q(x) + r$$

In particular, if $x=a$, this equation gives

$$\begin{aligned} P(a) &= (a-a)q(a) + r \\ &= r \end{aligned}$$

which proves theorem.

e.g. $\rightarrow$ Find the remainder when $x^4 + x^3 - 2x^2 + x + 1$ is divided by $x-1$

Solution: $\rightarrow$ Here $P(x) = x^4 + x^3 - 2x^2 + x + 1$ and zero of $x-1$ is 1

$$\begin{aligned} \text{So } P(1) &= (1)^4 + (1)^3 - 2(1)^2 + 1 + 1 \\ &= 2 \end{aligned}$$

So by remainder Theorem 2 is the remainder when the given $P(x)$ is divided by $(x-1)$

EXERCISE 2.3

⑥

Q1. Find the remainder when $x^3 + 3x^2 + 3x + 1$ is divided by
i) $x+1$

Sol: $\rightarrow$ Here $P(x) = x^3 + 3x^2 + 3x + 1$ and zero of $x+1 = -1$

$$\therefore P(-1) = (-1)^3 + 3(-1)^2 + 3(-1) + 1$$

$$= -1 + 3 - 3 + 1$$

$$= 0$$

$\therefore$ Remainder is 0

By division method : $\rightarrow$

$$\begin{array}{r} x+1 \overline{) x^3 + 3x^2 + 3x + 1} \quad (x^2 + 2x + 1) \\ \underline{x^3 + x^2} \\ 2x^2 + 3x + 1 \\ \underline{2x^2 + 2x} \\ x + 1 \\ \underline{x + 1} \\ 0 \end{array}$$

Hence remainder = 0

ii) $x - \frac{1}{2}$

Sol Here $P(x) = x^3 + 3x^2 + 3x + 1$ and zero of $x - \frac{1}{2}$ is $\frac{1}{2}$

$$P\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^3 + 3\left(\frac{1}{2}\right)^2 + 3\left(\frac{1}{2}\right) + 1$$

$$= \frac{1}{8} + \frac{3}{4} + \frac{3}{2} + 1 = \frac{1+6+12+8}{8} = \frac{27}{8} \text{ Ans.}$$